$\Lambda_b^0$ trigger

a)

PYTHIA 8.317; pp, $\sqrt{s}$ = 13.6 TeV
Hard $c\bar{c}$, $b\bar{b}$; $\hat{p}_T \geq 2$ GeV/c
$N_{ch}$: Charged primaries
$p_T > 0.15$ GeV/c; $|\eta| \leq 4$

MONASH
JUNCTIONS
CLOSEPACKING

b)

0-1%
40-50%
80-90%

$Y = \frac{N_{OS} - N_{SS}}{N_{trig}}$

c)

TUNE/MONASH

d)

$B^+$ $B^{*+}$ $B^0$ $B^{*0}$ $B_s^0$ $B_s^{*0}$ $B_c^+$ $B_c^{*+}$ $\bar{\Lambda}_b^0$ $\bar{\Sigma}_b^-$ $\bar{\Sigma}_b^{*-}$ $\bar{\Sigma}_b^0$ $\bar{\Sigma}_b^{*0}$ $\bar{\Sigma}_b^+$ $\bar{\Sigma}_b^{*+}$ $\bar{\Xi}_b^0$ $\bar{\Xi}_b^{'0}$ $\bar{\Xi}_b^{*0}$ $\bar{\Xi}_b^+$ $\bar{\Xi}_b^{'+}$ $\bar{\Xi}_b^{*+}$ $\bar{\Xi}_{bc}^-$ $\bar{\Xi}_{bc}^{'-}$ $\bar{\Xi}_{bc}^{*-}$ $\bar{\Xi}_{bc}^0$ $\bar{\Xi}_{bc}^{'0}$ $\bar{\Xi}_{bc}^{*0}$ $\bar{\Xi}_{bb}^0$ $\bar{\Xi}_{bb}^{*0}$ $\bar{\Xi}_{bb}^+$ $\bar{\Xi}_{bb}^{*+}$ $\bar{\Omega}_b^+$ $\bar{\Omega}_b^{*+}$ $\bar{\Omega}_{bc}^0$ $\bar{\Omega}_{bc}^{'0}$ $\bar{\Omega}_{bc}^{*0}$ $\bar{\Omega}_{bcc}^-$ $\bar{\Omega}_{bcc}^{*-}$ $\bar{\Omega}_{bb}^+$ $\bar{\Omega}_{bb}^{*+}$ $\bar{\Omega}_{bbc}^0$ $\bar{\Omega}_{bbc}^{*0}$ $\bar{\Omega}_{bbb}^{*+}$

Associate species